Scientific misconduct [edit]

According to the Office of Research Integrity (ORI), Potti engaged in scientific misconduct[4] while a cancer researcher at both Duke University's Medical Center and School of Medicine. He resigned in November 2010 after Duke put him on administrative leave, terminated the clinical trials based on his research and retracted his published data.[9][12][11]

Potti and his team were accused of falsifying data regarding the use of microarray genetic analysis for personalised cancer treatment, which was published in various prestigious scientific journals. While there were questions concerning Potti's work beginning in 2007, notably from two bioinformatic statisticians, Keith Baggerly and Kevin Coombes at MD Anderson Cancer Center,[13][14] 2010 brought further and more widespread scrutiny when it was discovered by Paul Goldberg and reported in The Cancer Letter[15][16] that Potti had claimed on his curriculum vitae that he had been a "Rhodes Scholar (Australian Board)".[12][17] He said he was referring to the Association of Rhodes Scholars in Australia Scholarships,[18] an award granted by an organisation of former Rhodes Scholars to bring Commonwealth citizens who attend overseas institutions in to Australia.[17][19][20]

Potti's fraudulent research was funded by the US government through the National Institutes of Health; the National Heart, Lung, and Blood Institute; and the National Cancer Institute, in the form of six multi-year grants,[21] and by the Howard Hughes Medical Institute.[22]

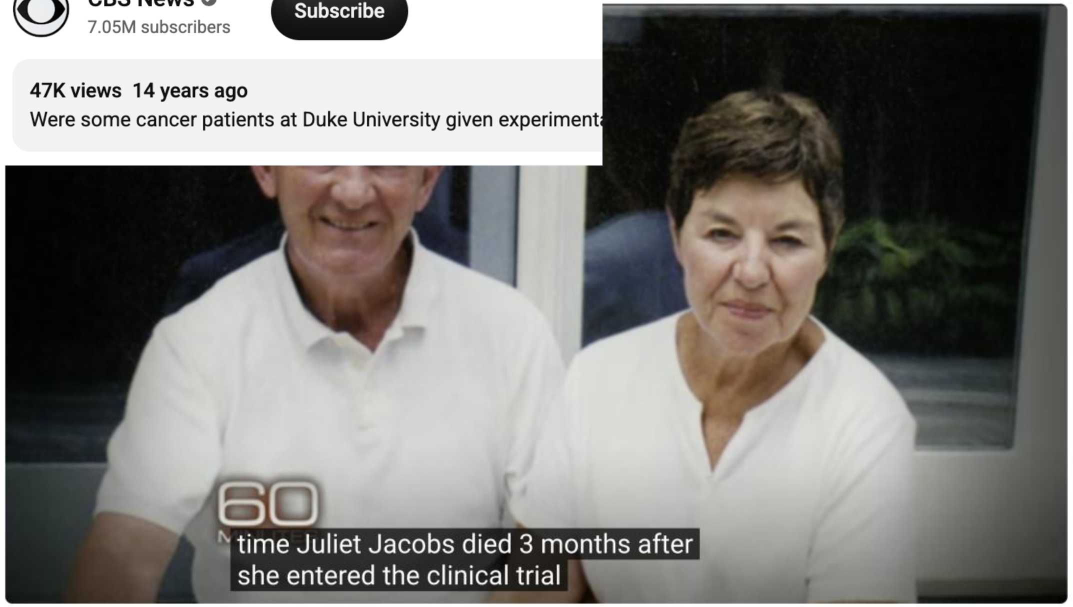
60 Minutes described the case as "one of the biggest medical research frauds ever".[23]

https://en.wikipedia.org/wiki/Anil_Potti

Deception at Duke

CBS News
7.05M subscribers
Subscribe

47K views 14 years ago
Were some cancer patients at Duke University given experiment



https://www.youtube.com/watch?v=eV9dcAGaVU8

NIH Investment in Independent AI Model Validation

Contact PI/Project Leader	Project Number	Awardee Organization	Areas of Research & ADSI Task Areas
Liu, Zhandong	1OT2OD040565	Baylor College of Medicine	Task IV: Model Validation/Replication

Validate ASD: Independent Multimodal Replication and Validation of Autism Data-Science Models

This project will develop a robust and transparent framework for the validation and replication of autism data science models through a multi-modal, cross-institutional approach. The initiative will employ comprehensive datasets from various sources, including clinical, genomic, environmental, and physiological data from Texas Children's Hospital (TCH) and other prominent research repositories. By leveraging advanced AI/ML methods, we will rigorously assess model performance across different pediatric populations, ensuring generalizability and fairness in clinical settings. We will implement a two-pronged validation strategy, including intact model testing and code-blinded replication. This methodology will rigorously evaluate the ac
predictive models. Key to this effort will be a
Warehouse, combined with genomic data fro
cohort and metabolomics data from the BaB
will be validated with placental biomarkers l

The project will provide essential insights int
applications, guiding their future clinical dep
stakeholder engagement, the project will de
that ensure models are utilized effectively ac
enhance autism care and set a new standard
research.

VALIDATE ASD: INDEPENDENT MULTIMODAL REPLICATION AND VALIDATION OF AUTISM DATA-SCIENCE MODELS

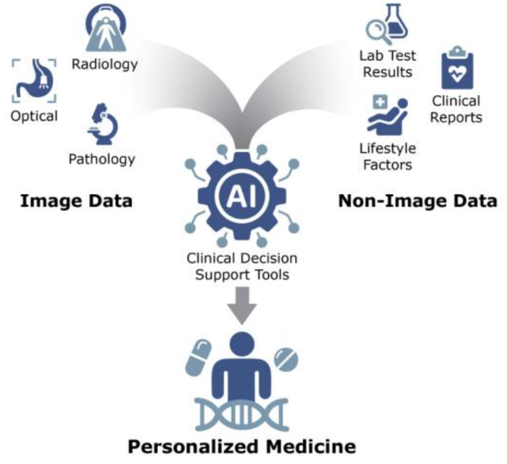
Project Number	Contact PI/Project Leader	Awardee Organization
1OT2OD040565-01	LIU, ZHANDONG Other PIs	

Details

Contact PI/ Project Leader	Other PIs	Program Official
Name LIU, ZHANDONG Title ASSOCIATE PROFESSOR Contact View Email	Name ANNAPRAGADA, ANANTH V HARRIS, HOLLY KATHRYN JEONG, HYUN-HWAN MOORTHY, BHAGAVATULA	Name KANO, KRISTIN KATHLEEN Contact View Email

Program Snapshot

The NIH Common Fund's **Precision Medicine with AI: Integrating Imaging with Multimodal Data (PRIMED-AI)** program will combine clinical imaging with other types of health data ("multimodal data") to develop innovative artificial intelligence (AI)-powered clinical decision support (CDS) tools to enable new personalized medicine strategies.



CDS tools are designed to help healthcare providers make informed decisions about a patient's care by

- Generating a validation framework for rigorous, impartial assessment of the overall dependability of AI models developed across the PRIMED-AI ecosystem and beyond.
- Supporting software tool development to address anticipated needs in key areas, as well as the development of standardized processes to support this emerging field.
- Enhancing communication between physicians, scientists, and patients within the PRIMED-AI space.

Verification Approaches (from ADSI validation tutorial)

- Four primary approaches to verifying research workflows and findings:

Reproduction	Validation	Robustness	Replication
Original Model Original Data New Hardware	Original Model New Data New Hardware	Recoded Model Original Data New Hardware	Recoded Model New Data New Hardware

Level of Rigor

Summary of Primary Verification Approaches

	Reproduction	Robustness	Validation	Replication
Question Addressed	Can the results be regenerated?	Does the finding hold under alternative methods?	Does the model work elsewhere?	Does the finding hold independently in a new dataset?
Data	Original Data	Original Data	**Internal:** Original Data *("Holdout")* **External:** New Data (*Different Population*)	New Data
Code/Model	Original Code	Alternative Model	Original Model (no refit)	Respecified Code
Focus	Computational Integrity	Sensitivity to Model Assumptions	Generalizability	Scientific Consistency
Success Metrics	Identical Output *(Estimates, Tables, Figures)*	Concordant results observed over a range of model/parameters	Model Performance Metrics (*Calibration, Discrimination, etc.)*	Consistent Magnitude and Direction of Results, Overlapping CIs
Common Failure Causes	Undocumented Steps, Code Errors, Computing Environment Variation	Misaligned methods with data distribution or baseline control models	**Internal:** Data Leakage, Overfitting **External:** Overfitting, Population Shift	Study-Specific Bias, Confounding, Insufficient Documentation

AURORA

Automated Understanding and Reproduction of Research Artifacts

- Brandon Lee

Project status update

What is AURORA?

One sentence

A pipeline that finds, filters, and (eventually) evaluates ASD research papers for genuine, reproducible data-science content.

Find

Pull the entire PMC open-access corpus in bulk

Filter

Keep only papers with genuine empirical, quantitative work

Evaluate

Score survivors on reproducibility (planned)

AURORA — Automated Understanding and Reproduction of Research Artifacts

Basic overview.

What

Build a clean, reproducible corpus of ASD papers that actually contain data-science / quantitative work — automatically, at the scale of millions of articles.

Why

Relevant ASD literature is scattered across ~8.8M open-access papers. Manual curation doesn't scale, and downstream analysis needs a filtered, trustworthy input set.

How

A staged funnel: download & index everything from S3, filter for relevance with regex + ML, then evaluate survivors for reproducibility.

One pipeline, three stages, cheapest filter first

Each stage narrows the corpus and is validated independently before the next runs.

STAGE 1

Download & Index

8.8M articles → local corpus

Bulk-pull text + metadata from the PMC S3 bucket; build a searchable index.

Complete

→

STAGE 2

Relevance Filter

ASD + data-science + primary

Regex pre-screen + ML classifier over full text; keep papers that do quantitative work.

In progress

→

STAGE 3

Reproducing

test the survivors

Take a paper's own data and methods and try to regenerate its reported result.

In progress

STAGE 1

Download & Index

Getting 8.8M papers onto disk, in bulk, from S3.

BACKGROUND

8.8M articles, one static corpus to get onto disk

The source

PMC's entire open-access corpus lives on a public AWS S3 bucket — no credentials required. Before anything can be filtered or scored, it has to be pulled down and indexed once.

8.8M

articles total

~0.27 TB

plain text

4.8 TB

full PDFs (skipped)

ORIGINAL PLAN

Three access paths, combined

OA File List + E-utilities + BioC

- **OA File List (manifest)**
  Enumerate the corpus up front, filter by license/date.
- **E-utilities (esearch / efetch)**
  Keyword / MeSH search to find ASD papers.
- **BioC API**
  Pull structured full text for matched articles.

The idea: use all three together, so ASD-relevance filtering happens at fetch time instead of afterward.

WHAT WE TRIED

EVIDENCE

Too slow at corpus scale — the numbers behind the switch

Two problems ruled the API out: 3 req/s with no bulk fetcher, and recall risk from keyword-only search. The comparison below is why bulk S3 won.

	esearch / efetch	S3 bulk download
Method	Live API query	One-time static manifest
Rate limit	3 req/s (unauthenticated)	— (public bucket)
Throughput	theoretical only — no bulk fetcher	measured ~16 papers/sec

Of the 1,000 pilot papers: 820 got full text, 180 metadata-only (no DOI), 0 failures. Bulk S3 is the ingestion backbone; the API now only pulls small PMID sets for ground truth.

WHAT WE'RE DOING

Bulk S3 download, with resumable state

1. Read the S3 inventory once

a static manifest of every article — no live per-record calls.

2. Shard + download in parallel

text + metadata pulled from S3, MD5-verified per file.

3. Index into DuckDB

one searchable metadata table over the whole corpus.

- **SQLite (WAL) for download state** — resumable across 32 workers
- **DuckDB for the metadata index** — fast search over millions of rows
- **Latest version only** — v2 supersedes v1 where both exist
- **Per-job shard DBs** — DuckDB is single-writer; merge after
- **PMCID-prefix directories** — avoids inode exhaustion at 8.8M files
- **Idempotent, per-article commits** — resilient to Slurm job kills

STAGE 2

Relevance Filter

Which papers actually do the empirical, quantitative work?

BACKGROUND

Mentioning a topic isn't the same as studying it

The problem

A regex or keyword hit tells you a word appeared — not whether the paper actually did that work. A review that cites twelve ML papers and a paper that runs its own model both trip the same keyword.

What Stage 2 has to decide

Given a paper already matched as ASD-related: does it contain genuine empirical / data-science content worth carrying forward, or should it be dropped (case report, narrative review, opinion piece)?

ORIGINAL PLAN

The original plan: two filters, ASD and empirical

ASD-focused

Autism / ASD is the actual subject of the study.

Empirical / data-science

Reports a real quantitative result, or uses genuine ML / statistical methods.

Original tooling

Both criteria checked by regex or FastText.

Question

Is an LLM the answer?

The number of papers

Corpus size: roughly 8.8 million articles in the PMC open-access corpus — far beyond what per-document LLM inference can affordably cover.

Even a small subset adds up: ~50k ASD article subset would be roughly $500 with Haiku4.5,

EVIDENCE

What is TF-IDF?

Term Frequency – Inverse Document Frequency: a literal-vocabulary matching score.

How it works

Term Frequency: how often a word appears in this document.

Inverse Document Frequency: down-weight common words (“the”), up-weight rare, distinctive ones.

Result: each document becomes a weighted word-count vector; logistic regression learns which words separate keep from exclude.

Why it wins here

- **No token limit —**

whole articles in, no chunking.

- **Fast, CPU-only —**

minutes for the whole corpus.

- **Interpretable —**

you can see which words drove a decision.

- **Limitation:** only catches literal wording — a paraphrased paper can slip past.

EVIDENCE

What is a MiniLM embedding?

A compact transformer that turns text into a meaning vector, not a word-count vector.

How it works

all-MiniLM-L6-v2 outputs a 384-dim vector meant to capture *meaning*, not just which words appeared.

The promise: two papers describing the same work in different words should land near each other — something TF-IDF can't do.

The catch

- **256-token limit**

— forces chunking on full text.

- **Opaque dimensions —**

can't inspect why it decided something.

- **Underperforms TF-IDF —**

on abstracts, full text, and real unlabeled data (later slides).

Question

How do we get our benchmark data?

Why abstracts, why ASD papers

E-utilities makes it easy to pull abstracts for ASD papers specifically.

- Assumption training data should look like the target domain

How labeling worked

Manually labeled abstracts.

The realization that reshaped the ground truth

The realization

ASD is already handled by the regex filter (previous slides). This classifier only has to answer one question: does the text describe genuine ML / data-science work? Training text doesn't need to be ASD-specific at all.

Why it matters for testing

Testing this classifier on off-topic, non-ASD articles (Pilot2, later slides) is a valid, independent check — not a mismatch. Since ASD relevance is regex's job, a fair test of this classifier is whether it can spot empirical content on any subject.

EVIDENCE

Abstracts (Run 1): TF-IDF + Random Forest edges it out

234 examples (107 keep / 127 exclude), 5-fold CV. Abstracts were tested first.

Representation	Classifier	F1	± std	Precision	Recall
TF-IDF	**RandomForest**	**0.883**	**0.019**	**0.921**	**0.851**
TF-IDF	LogisticRegression	0.876	0.030	0.928	0.833
TF-IDF	ComplementNB	0.871	0.027	0.927	0.823
MiniLM embeddings	LinearSVC	0.865	0.056	0.897	0.840
TF-IDF	LinearSVC	0.864	0.039	0.936	0.805
MiniLM embeddings	LogisticRegression	0.827	0.042	0.879	0.784
TF-IDF	XGBoost	0.817	0.063	0.874	0.775
MiniLM embeddings	XGBoost	0.798	0.033	0.825	0.775
MiniLM embeddings	RandomForest	0.783	0.057	0.871	0.719
MiniLM embeddings	ComplementNB	0.782	0.077	0.848	0.729

Statistically, RandomForest and LogReg are indistinguishable here (within 1 pooled std) — LogReg is kept as the reference point because it stays strongest once full text is added.

EVIDENCE

Full text (Run 2): TF-IDF + Logistic Regression wins clearly

107 examples (50 keep / 57 exclude), 5-fold CV, identical fold splits to the abstract run.

Representation	Classifier	F1	± std	Precision	Recall
TF-IDF	**LogisticRegression**	**0.921**	**0.049**	**0.924**	**0.920**
TF-IDF	ComplementNB	0.901	0.069	0.904	0.900
TF-IDF	LinearSVC	0.876	0.063	0.905	0.860
TF-IDF	RandomForest	0.862	0.098	0.848	0.880
MiniLM embeddings	LinearSVC	0.858	0.036	0.881	0.840
MiniLM embeddings	LogisticRegression	0.844	0.030	0.836	0.860
MiniLM embeddings	RandomForest	0.840	0.079	0.805	0.880
TF-IDF	XGBoost	0.818	0.048	0.906	0.760
MiniLM embeddings	ComplementNB	0.808	0.069	0.802	0.820
MiniLM embeddings	XGBoost	0.718	0.061	0.762	0.680

TF-IDF fills 4 of the top 4 spots. Full text lifts LogReg's F1 by +0.045 over abstracts — richer text helps the winning model most.

EVIDENCE

Confusion matrices: full text has zero false negatives

80/20 held-out split, 107-paper matched subset, identical papers across all 4 representations.

A: Abstract TF-IDF

10 TN	2 FP
1 FN	9 TP

B: Full-text TF-IDF

9 TN	3 FP
0 FN	10 TP

C: MiniLM MAX-pool

8 TN	4 FP
3 FN	7 TP

D: MiniLM MEAN-pool

8 TN	4 FP
2 FN	8 TP

Reading it

Full-text TF-IDF (B) misses zero real keep-worthy papers on this split — the best recall on the metric that matters most, since missing a real paper costs more than including a marginal one. Both MiniLM poolings (C, D) are weakest overall.

EVIDENCE

Pilot2: testing the classifier on 673 real, unlabeled articles

What it is

673 real PMC articles, pulled in bulk with zero ASD filtering.

Why off-topic is the point: ASD relevance is already regex's job (previous slides), so this classifier only has to catch empirical content.

What the regex screen found

482 / 673 (71.6%) of these off-topic articles matched a named-method or quantitative-result pattern — a first-pass estimate of how much of Pilot2 is genuinely empirical.

EVIDENCE

Full sweep on Pilot2

All 10 combinations scored the same 673 articles — keep/exclude counts and keep rate by representation and classifier.

Representation	Classifier	Keep	Exclude	Keep rate
TF-IDF	LogisticRegression	565	108	**84.0%**
TF-IDF	LinearSVC	571	102	**84.8%**
TF-IDF	ComplementNB	534	139	**79.3%**
TF-IDF	XGBoost	580	93	**86.2%**
TF-IDF	RandomForest	574	99	**85.3%**
MiniLM	LogisticRegression	327	346	**48.6%**
MiniLM	LinearSVC	404	269	**60.0%**
MiniLM	ComplementNB	270	403	**40.1%**
MiniLM	XGBoost	254	419	**37.7%**
MiniLM	RandomForest	229	444	**34.0%**

Striking split: every TF-IDF combo keeps 79–86%, every MiniLM combo keeps 34–60%. Representation matters far more than classifier.

WHAT WE'RE DOING

What we've settled on so far

Decision rule: regex + 2010 cutoff for ASD/reviews, then a context-aware classifier decides empirical relevance.

- **Ground truth: 274 examples**
- **Full text over abstracts** — +0.06 F1; training text no longer needs to be ASD-specific.
- **Reference stripper: built + tested**
- **TF-IDF + LogReg leads the sweep** — Best, most stable model on both abstracts and full text.

STAGE 3

Reproducing

Can we take a paper's own data and regenerate its result?

BACKGROUND

“Reproducible” has to mean something testable

The question Stage 3 has to answer

A paper that survives Stage 2 has real empirical content — but does it actually hold up? Can someone else take its stated data and methods and get the same result?

Why this is the hard stage

Stage 1 and 2 are about finding and filtering text. Stage 3 means actually running someone else's analysis — downloading their data, following their methods, and checking whether the numbers match.

ORIGINAL PLAN

A cheap rubric, scored across every paper

Data availability

Accession IDs, repository links, availability statements

Figure / table count

Density as a rough evidence proxy

Method clarity

Named methods + quantitative markers in Methods

Code / supplement

Mentions of shared code, scripts, supplements

The idea

Score every surviving paper on these four axes cheaply, with regex pattern-groups — a proxy for reproducibility across the whole corpus, without actually running anything.

What we've done so far

Stage 1: bulk S3 download built and piloted — 8.8M articles inventoried, 1,000-paper pilot, 0 failures.

Stage 2 filter chain designed and built — Regex ASD/review/2010 filter → context-aware empirical classifier.

Ground truth grown and cleaned — 10 → 274 examples, hard negatives added, positive-only bias fixed.

Full model sweep completed twice — 5 classifiers × 2 representations, on abstracts and full text.

Tested on real, unlabeled, off-topic data — 673-article pilot2 run — first genuine production-style test.

Next steps

In priority order.

- **Manually review the top pilot2 disagreements** — 20 cases where TF-IDF and MiniLM split — converts consistency into an actual accuracy check.
- **Re-validate ground truth against full text** — Close the train/inference mismatch — abstracts-labeled, full-text-inferred.
- **Validate the Data Availability guard at scale** — Built and unit-tested; confirm it holds across every journal's formatting quirks.
- **Decide MiniLM's fate** — It underperforms and under-scores on every test so far; keep only if ground truth grows to include real paraphrased cases.
- **Run Stage 2 on the full candidate pool** — Write relevance_label / relevance_score back to DuckDB with TF-IDF + LogReg.
- **Begin Stage 3 implementation** — Reconcile the reproducibility rubric concept with the existing single-paper pilot.

Acknowledgements

Hwan – Zach – Kelly – Jeong Lab

Any Questions?